

JKBOSE 2026

CLASS 11TH

BIOLOGICAL CLASSIFICATION

LECTURE-01

BOTANY

ARSHID BHAT SIR



Topics To Covered

- 1** Introduction ✓
- 2** Kingdom Classification of Living Organisms ✓

INTRODUCTION

≡ Biological Classification — It is a process by which living organisms are classified into various categories on the basis of some easily observable characters

≡ Purpose of Classification — To make study of vast no. of living organisms easy.

Aristotle — Was the first person who Scientifically Classified living organisms on the basis of some Simple morphological Characters.

Aristotle — Classified plants into 03 groups (on the basis of habit)

a. Herbs ✓

b. Shrubs ✓

c. Trees ✓

Aristotle — Classified animals into two groups.

a. Enaima (posses red blood)

b. Anaima (does not possess red blood)

Without.

NCERT READING



Since the dawn of civilisation, there have been many attempts to classify living organisms. It was done instinctively not using criteria that were scientific but borne out of a need to use organisms for our own use – for food, shelter and clothing. Aristotle was the earliest to attempt a more scientific basis for classification. He used simple morphological characters to classify plants into trees, shrubs and herbs. He also divided animals into two groups, those which had red blood and those that did not.

KINGDOM CLASSIFICATION OF LIVING ORGANISMS

- ① Two Kingdom Classification. ✓
- ② Three " " "
- ③ Four " " "
- ④ Five " " "
- ⑤ Six " " "

TWO KINGDOM CLASSIFICATION

- ≡ Introduced by Carolus Linnaeus
- ≡ Used till (before 1969).
- ≡ He divided all the living organisms into two Kingdoms:
 - a. Kingdom Plantae.
 - b. Kingdom Animalia.

All living organisms

1. Kingdom Plantae

- Algae
 - Chlamydomonas (unicellular, motile)
 - Spirogyra (multicellular)
 - Chlorella (unicellular, non-motile)
 - Bryophyta, ~~Pteridophyta~~, ~~Gymnosperms~~, ~~Angiosperms~~
 - Cyanobacteria (Blue green algae)
 - Archaeobacteria
 - Fungi (heterotrophic)
- } prokaryotes

2. Kingdom Animalia

- Mammals ✓
- Aves (Birds) ✓
- Reptiles ✓
- Fishes (pisces) ✓
- Protozoans (Amoeba, Paramecium)
- Vertebrates
- Invertebrates

Criteria followed :-

- ① Cell wall
 - Present in plants
 - Absent in animals.
- ② Locomotion
 - Absent in plants
 - Present in animals.
- ③ Mode of Nutrition
 - Plants are autotrophic.
 - Animals are heterotrophic.

Drawbacks :-

① Unicellular & multicellular organisms placed together.

Ex. Spirogyra (multicellular)

Chlamydomonas (unicellular)

} Placed in Plantae
under Algae

② Prokaryotes and Eukaryotes — placed together.

Ex. Cyanobacteria (prokaryote) — placed with Eukaryotes.

③. Autotrophs and Heterotrophs — placed together.

Ex. Fungi (heterotrophic) — placed in plants (Autotrophic).

④. Position of Some organisms — was Contraversial.

Ex. Chlamydomonas { motile (Animal like character)
Autotrophic (Plant like character)

Note:- Understanding of classification increases over time.

NCERT READING



In Linnaeus' time a **Two Kingdom** system of classification with **Plantae** and **Animalia** kingdoms was developed that included all plants and animals respectively. This system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and photosynthetic (green algae) and non-photosynthetic (fungi) organisms. Classification of organisms into plants and animals was easily done and was easy to understand, but, a large number of organisms did not fall into either category. Hence the two kingdom classification used for a long time was found inadequate. Besides, gross morphology a need was also felt for including other characteristics like cell structure, nature of wall, mode of nutrition, habitat, methods of reproduction, evolutionary relationships, etc. Classification systems for the living organisms have hence, undergone several changes over the time. Though plant and animal kingdoms have been a constant under all different systems, the understanding of what groups/organisms be included under these kingdoms have been changing; the number and nature of other kingdoms have also been understood differently by different scientists over the time.

THREE KINGDOM CLASSIFICATION

- Introduced by Ernest Haeckel.

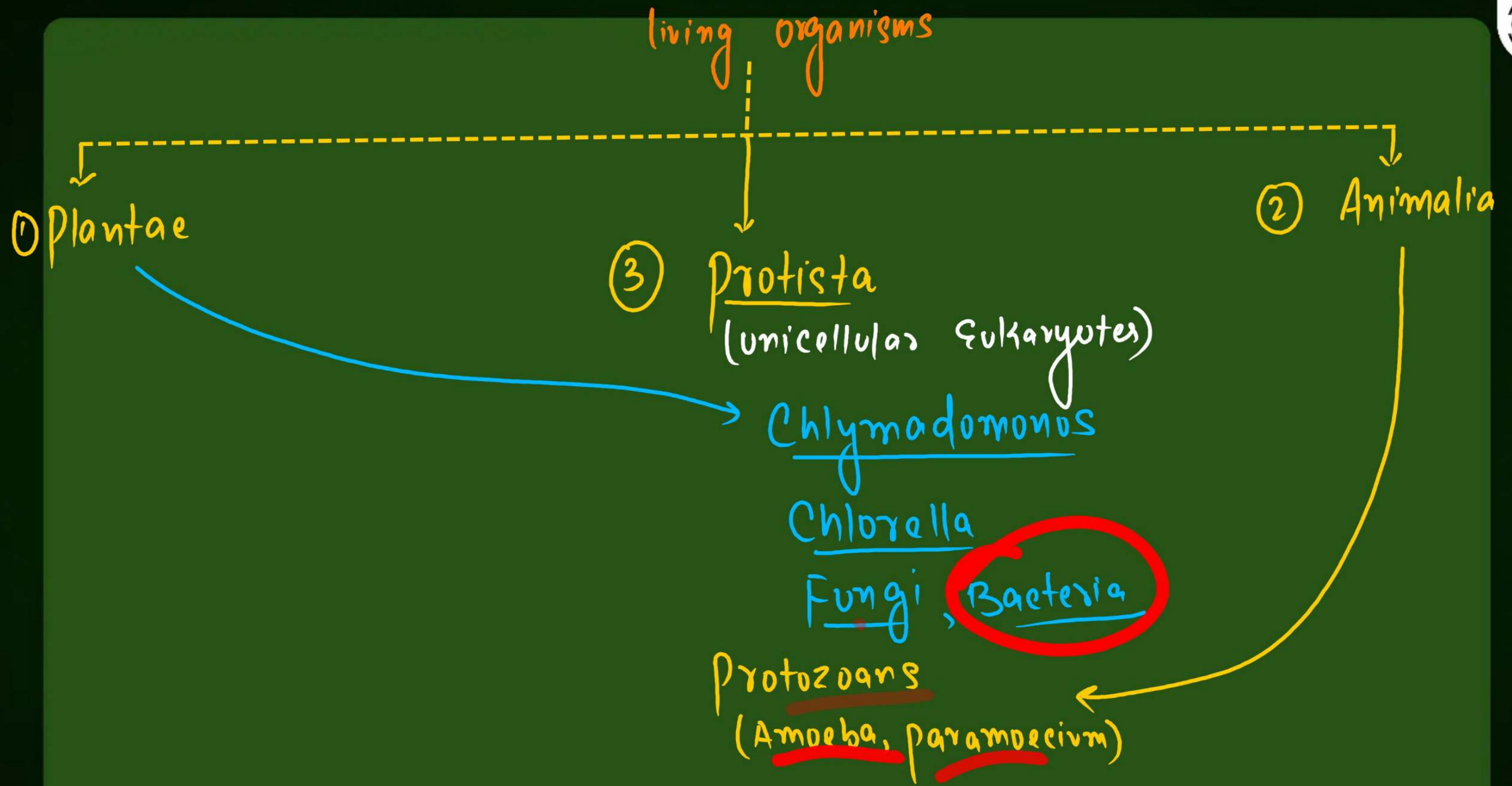
- He divided living organisms into

Kingdom Plantae

Kingdom Animalia

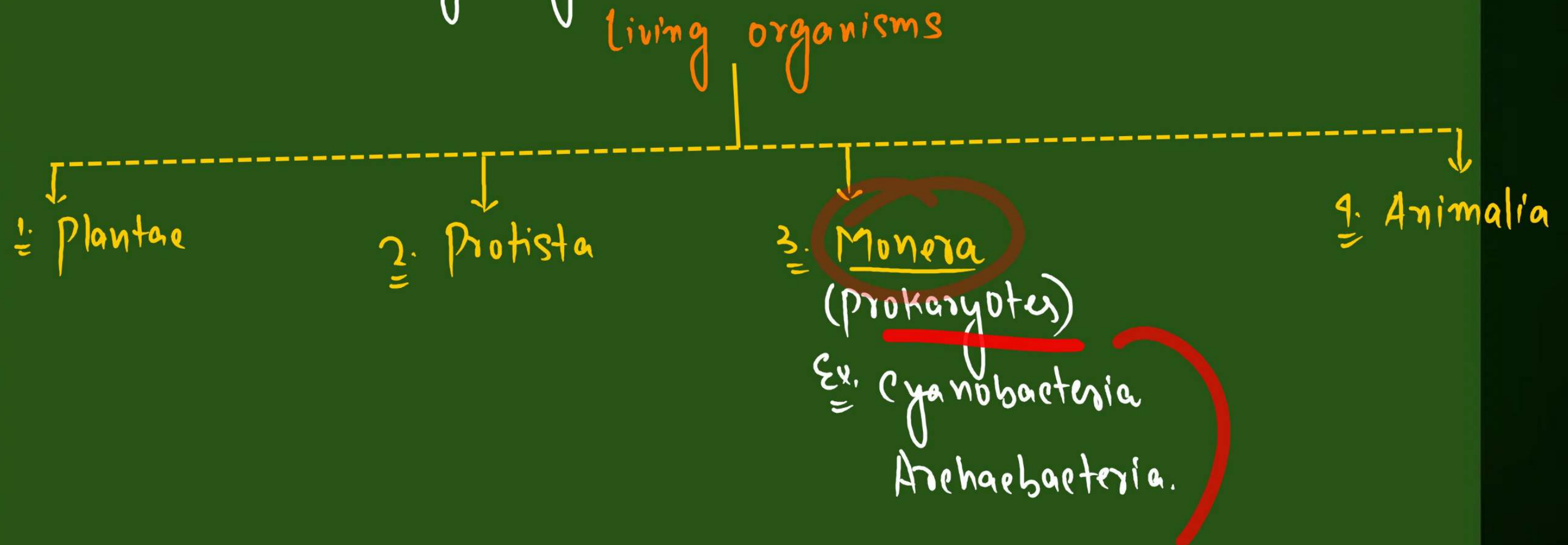
Kingdom Protista.

- Kingdom Protista includes — all "unicellular Eukaryotes"



FOUR KINGDOM CLASSIFICATION

- Introduced by Copeland.
- He divided living organisms into 04 kingdoms.

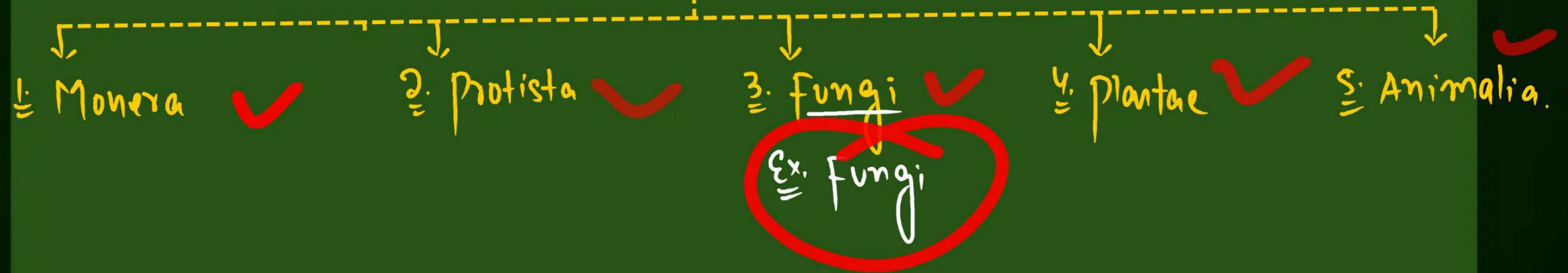


FIVE KINGDOM CLASSIFICATION

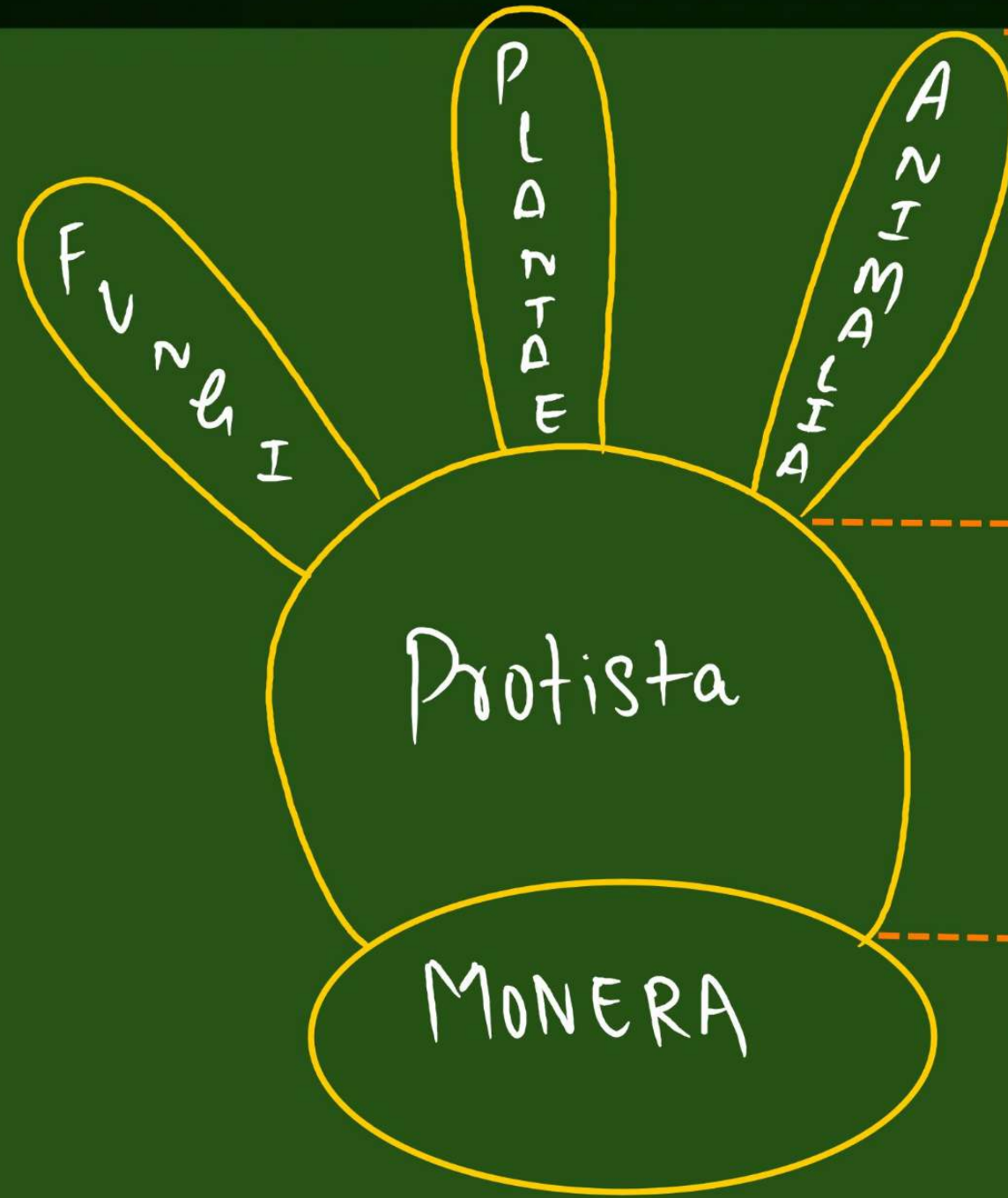
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- Introduced by R. H. Whittaker (1969)
- He divided living organisms into 5 Kingdoms:

living organisms



Direction of Evolution



- Eukaryotes
- Multicellular.

- Eukaryotes
- Unicellular

- prokaryotes
- unicellular

Criteria followed :-

- ① Complexity of Cell Structure (prokaryotic / eukaryotic).
- ② Body Organisation. (unicellular / multicellular).
- ③ Mode of Nutrition (Autotrophic / Heterotrophic).
- ④ Mode of reproduction. (Asexual / Sexual).
- ⑤ Evolutionary (Phylogenetic) relationship among organisms.

Drawbacks :-

- ① No place for — viruses, viroids & prions.
- ② No place for Lichen (symbiotic association b/w algae and fungi).

Note:- As per 5-kingdom classification, Chlamydomonas, Chlorella are placed in Protista Kingdom.

SIX KINGDOM CLASSIFICATION



9th NEXT lecture
??

H/w.

QUESTIONS

Five kingdom system of classification suggested by R.H. Whittaker is not based on [MR* 2014]

- 1 Complexity of body organization
- 2 Presence or absence of a well defined nucleus
- 3 Mode of reproduction
- 4 Mode of nutrition

QUESTIONS

In the five-kingdom classification, *Chlamydomonas* and *Chlorella* have been include in: [2012 Mains]

- ☒ 1 Protista
- ☐ 2 Algae
- ☐ 3 Plantae
- ☐ 4 Monera

QUESTIONS

H/w

In five kingdom system, the main basis of classification:

[2002]

- 1 Structure of nucleus
- 2 Mode of nutrition
- 3 Structure of cell wall
- 4 Asexual reproduction

HOMEWORK



2 Questions

Thank
You